Date: Oct 27, 2025
To: "Hector Hito Sepulveda" hectorsepulveda@udec.cl
From: "Ocean Modelling" noreply_EMsupport@elsevier.com
Subject: Decision on submission to Ocean Modelling

Manuscript Number: **OCEMOD-D-25-00208**

Origin and dynamics of the Gulf of Ancud's eddy, in Chile: A numerical analysis

Dear Ph.D. Sepulveda,

Thank you for submitting your manuscript to Ocean Modelling.

I have completed my evaluation of your manuscript. The reviewers recommend reconsideration of your manuscript following major revision. I invite you to resubmit your manuscript after addressing the comments below. Please resubmit your revised manuscript by **Nov 24, 2025**.

When revising your manuscript, please consider all issues mentioned in the reviewers' comments carefully: please outline every change made in response to their comments and provide suitable rebuttals for any comments not addressed. Please note that your revised submission may need to be re-reviewed.

To submit your revised manuscript, please log in as an author at https://www.editorialmanager.com/ocemod/, and navigate to the "Submissions Needing Revision" folder.

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Ocean Modelling values your contribution and I look forward to receiving your revised manuscript.

Kind regards,
Changming Dong
Associate Editor

Ocean Modelling

Editor and Reviewer comments:

Reviewer's Responses to Questions

Note: In order to effectively convey your recommendations for improvement to the author(s), and help editors make well-informed and efficient decisions, we ask you to answer the following specific questions about the manuscript and provide additional suggestions where appropriate.

1. Are the objectives and the rationale of the study clearly stated?

Please provide suggestions to the author(s) on how to improve the clarity of the objectives and rationale of the study. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: The goal of the numerical modeling work is to study the origin and dynamics of the eddy in the Gulf of Ancud. Sensitivity experiments are performed. However, the original and dynamics of the eddy are still not clear. The main conclusion is that tide forcing does not play a crucial role in the generation of the eddies. Rather, those eddies are vaguely claimed to be robust elements of the local dynamics. Nothing is really convincing at all.

Reviewer #2: Mostly yes, but I have a major comment.
1. The focus on this study is mesoscale eddies in the Gulf of Ancud. But I feel that the eddies (shown in Figure 11-12) could fall in the submesoscale regime since their diameters are 10-30 km. Submesoscale eddies are generally affected by dynamics different from mesoscale, e.g., mixed layer instabilities, and would change the physical explanations of the results. Do the authors agree that the eddies they detect could fall in the submesoscale range?

2. If applicable, is the application/theory/method/study reported in sufficient detail to allow for its replicability and/or reproducibility?

Please provide suggestions to the author(s) on how to improve the replicability/reproducibility of their study. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: Mark as appropriate with an X:
Yes [] No [] N/A [x]
Provide further comments here:

Reviewer #2: Mark as appropriate with an X:
Yes [X] No [] N/A []
Provide further comments here:

Mostly yes.
1. Why choose 150 m (Line 491) as the constant depth in one of the sensitivity experiments. Is 150 m related to mixed layer depth?

2. How are the climatological winds defined? Is it time-invariant over the whole month of Jan/July? Or does it vary over the month?

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3. If applicable, are statistical analyses, controls, sampling mechanism, and statistical reporting (e.g., P-values, CIs, effect sizes) appropriate and well described?

Please clearly indicate if the manuscript requires additional peer review by a statistician. Kindly provide suggestions to the author(s) on how to improve the statistical analyses, controls, sampling mechanism, or statistical reporting. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: Mark as appropriate with an X:
Yes [] No [] N/A [x]
Provide further comments here:

Reviewer #2: Mark as appropriate with an X:
Yes [] No [] N/A [X]
Provide further comments here:

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4. Could the manuscript benefit from additional tables or figures, or from improving or removing (some of the) existing ones?

Please provide specific suggestions for improvements, removals, or additions of figures or tables. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: no

Reviewer #2: - Figure 1 is too general. I suggest integrating it into Figure 3, e.g., adding a subplot showing the global location of the Gulf of Ancud in Figure 3. Simply removing Figure 1 is also fine.

- Figure 2 is not directly relevant since it tells global distribution of fjord, which is not the focus of this study. I suggest removing it.

- Figure 5 is from other study. I suggest removing it since the whole point is just to show that there exist circular motions in the Gulf of Ancud. This is already done by citing Allel (2020) in text.

- I do not quite understand why showing Figures 7 and 8. To me, they are telling the same thing: wind is southeastward (northwest) in winter (summer) in the studied region. Also, this is not an evaluation of the model since the model is ocean only and the wind is from data, which shall be already reported in previous studies. Could the authors add more explanation?

- Figure 11 & 12: Could the authors move labels (a) to (h) close to the subplots they are referring to. For example, "(a) Control case" had better to be top of the top left plot. Also, are the velocities monthly averaged? Please clarify in the caption.

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5. If applicable, are the interpretation of results and study conclusions supported by the data?

Please provide suggestions (if needed) to the author(s) on how to improve, tone down, or expand the study interpretations/conclusions. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: Mark as appropriate with an X:
Yes [] No [] N/A [x]
Provide further comments here:

Reviewer #2: Mark as appropriate with an X:
Yes [] No [] N/A []

Provide further comments here:

1. Line 658-663: Figure 10b does not show any clear spatial patterns, since the field is highly homogeneous in space.

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6. Have the authors clearly emphasized the strengths of their study/theory/methods/argument?

Please provide suggestions to the author(s) on how to better emphasize the strengths of their study. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: no

Reviewer #2: Mostly yes.

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7. Have the authors clearly stated the limitations of their study/theory/methods/argument?

Please list the limitations that the author(s) need to add or emphasize. Please number each limitation so that author(s) can more easily respond.

Reviewer #1: no

Reviewer #2: No.
The main approach of this study is sensitivity experiments. However, all simulations were conducted for only two months (January and July), and the analyses are entirely based on the outputs from these limited periods. I am concerned that such short simulations may not be sufficient to yield statistically robust results for the eddy count. As shown in Table 2, the number of eddies identified in each experiment is only 2–4, which raises the question of whether these counts would diverge in longer simulations. This is a notable limitation of the study, and I would appreciate a more detailed explanation and/or an extension of the simulations to address this concern.

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8. Does the manuscript structure, flow or writing need improving (e.g., the addition of subheadings, shortening of text, reorganization of sections, or moving details from one section to another)?

Please provide suggestions to the author(s) on how to improve the manuscript structure and flow. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: The manuscript is poorly written with a lengthy introduction that tries to review ocean eddies but does a poor job. The content is poorly organized. Most of the paragraphs in the introduction can be simply removed.

Reviewer #2: Yes.
1. The Introduction is lengthy and includes background information that is not closely relevant to the main focus of the study. While I appreciate the authors' effort to provide a comprehensive context, the current version reads more like a dissertation than a research article. Moreover, the main message emerges too late — e.g., the central topic about eddies in the Gulf of Ancud only becomes clear around Line 154. I suggest substantially shortening the Introduction to make the study's objectives more apparent early on.

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9. Could the manuscript benefit from language editing?

Reviewer #1: Yes

Reviewer #2: No

Reviewer #1: Review of "Origina and dynamics of the Gulf of Ancud's eddy, in Chile: A numerical analysis" by Valderrama et al.

The goal of the numerical modeling work is to study the origin and dynamics of the eddy in the Gulf of Ancud. Sensitivity experiments are performed. However, the original and dynamics of the eddy are still not clear. The main conclusion is that tide forcing does not play a crucial role in the generation of the eddies. Rather, those eddies are vaguely claimed to be robust elements of the local dynamics. Nothing is really convincing at all.

The manuscript is poorly written with a lengthy introduction that tries to review ocean eddies but does a poor job. The content is poorly organized. Most of the paragraphs in the introduction can be simply removed.

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Reviewer #2: This study uses sensitivity experiments in a regional ocean model to investigate the forcings responsible for the generation of eddies in the Gulf of Ancud. The authors find that eddies consistently emerge in simulations without tidal forcing, with a flat bottom, and under climatological winds. They conclude that the Gulf of Ancud eddies are robust features of the local dynamics and are not significantly influenced by these forcings.
My major concerns about this study are twofold. First, it is unclear whether the simulation length is sufficient to yield

statistically robust eddy counts. Second, the dynamical regime of the eddies—whether they are submesoscale or mesoscale—requires clearer discussion, as it may affect the physical interpretation of the sensitivity experiments. Both points have been elaborated above. These issues, together with the other comments, should be addressed before the manuscript can be considered for publication.

I also list a few editorial comments below:
Line 87: "first" -> "upper"
Line 608: "velocity produced by the tide" -> "tidal currents"
Line 654-655: "residual tide" repeated as it is already defined in Line 638

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